

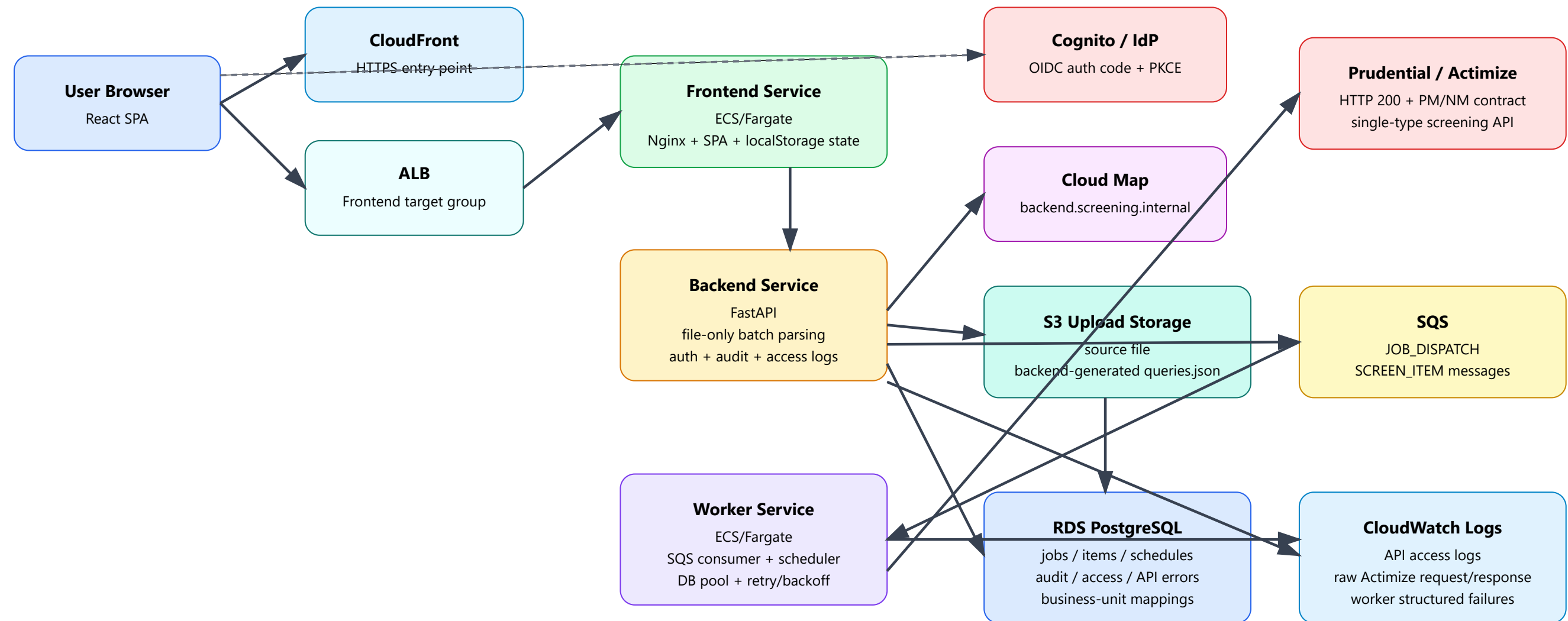
OFAC Screening Enterprise - Architecture and API Flow

Updated for the current implementation on 2026-04-09.

This graphical document is designed specifically for PDF export. Each page contains a rendered SVG diagram so the deployed architecture, API flow, and database relationships remain visible without requiring Mermaid support in the PDF viewer.

1) Deployed Architecture

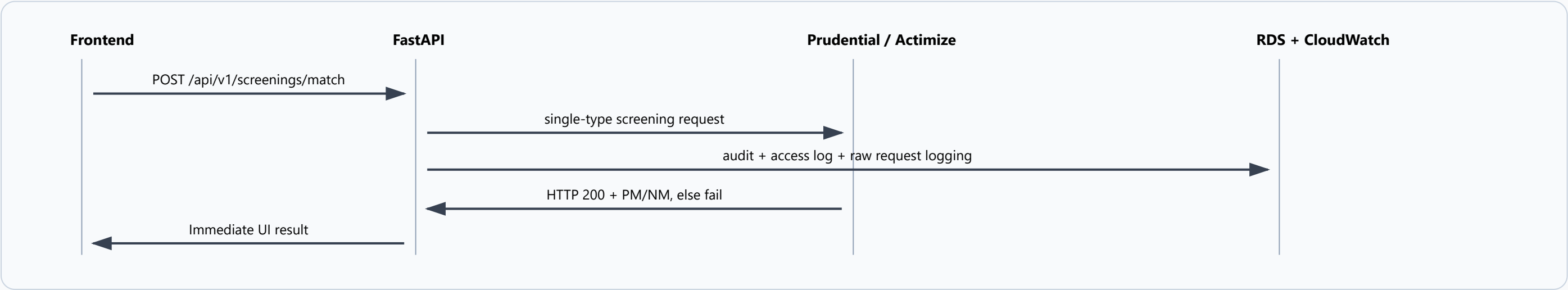
AWS Account / Environment



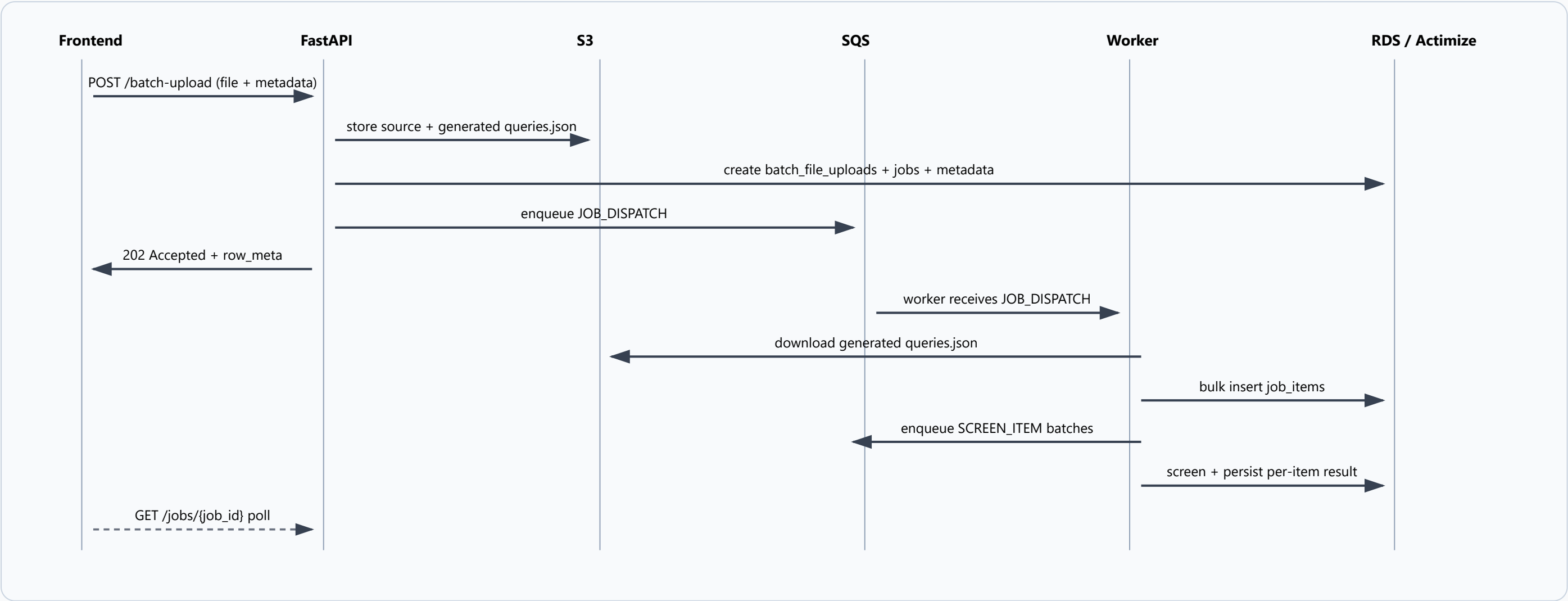
Browser traffic enters through CloudFront and ALB, reaches the frontend task, and `/api/\*` is reverse-proxied to FastAPI through Cloud Map. Immediate large batch uploads are now file-only from the browser; the backend parses the file, stores the source plus generated `queries.json`, and dispatches async work through SQS.

2) API Flow Patterns

Single Screening (Sync)

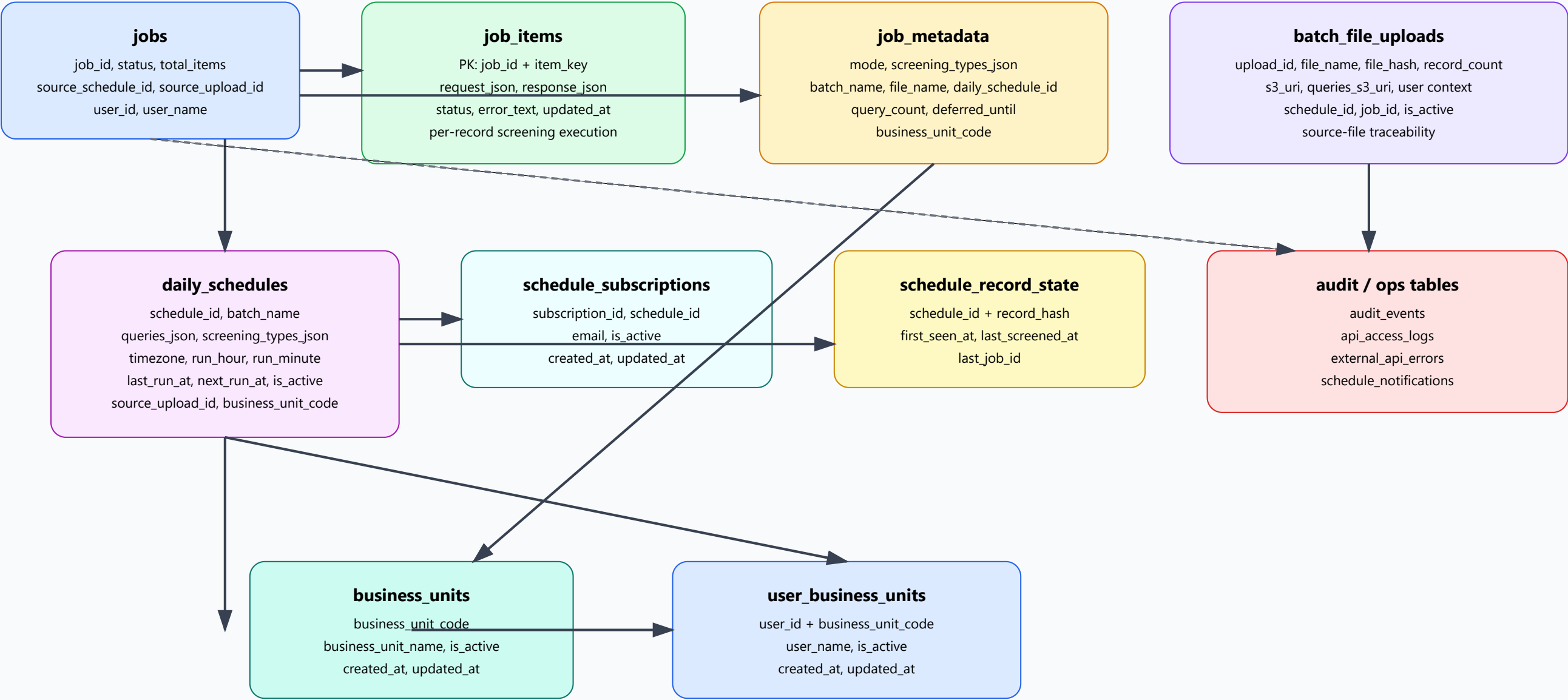


Batch Upload (Async, File-Only Upload Contract)



The browser no longer sends `queries.json`. FastAPI parses the uploaded file, validates fields such as `PartyKey`, gender, and names, generates normalized screening payloads, and only then dispatches async work. Under load, the client can still time out at the gateway while the backend completes the job creation path, so job existence must be confirmed server-side for support cases.

3) Database Design (Core Tables)



The execution core is `jobs` plus `job\_items`, with `job\_metadata` carrying UI/history fields. Scheduled runs, uploads, business-unit authorization, and audit/ops telemetry are stored in dedicated tables so each concern can be queried and retained independently.